REGIM DINAMIC

- Aprox Semnal Mic -

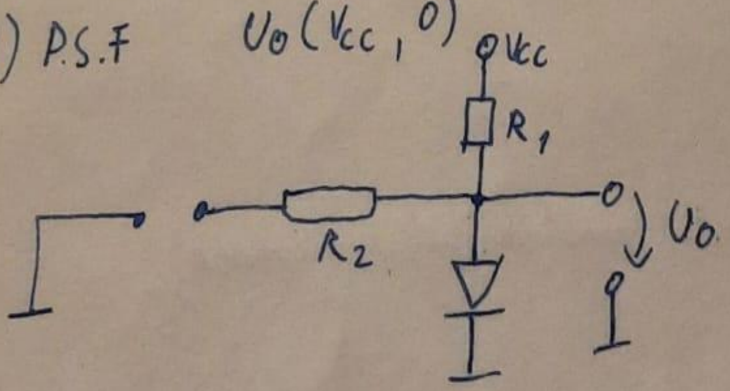
1) Calcul P.S.F. $U_0(V_{cc}, 0)$
2) Liniarizare în jurul P.S.F
3) Calculăm $U_0(0, U_i)$ cu circuitul liniarizat
4) Aplicăm Superpoziția $U_0(V_{cc}, U_i) = U_0(V_{cc}, 0) + U_0(0, U_i)$

$$I_D = I_S\left(e^{\frac{q_0 U_D}{KT}} - 1\right) \simeq I_S e^{\frac{q_0 U_D}{KT}}$$

$$\frac{dI_0}{dU_0} = \frac{d\left(I_S e^{\frac{q_0 U_D}{KT}}\right)}{dU_D} = \frac{q_0}{KT} I_S e^{\frac{q_0 U_0}{KT}} = \frac{q_0}{KT} I_D$$

$$R_D = \frac{dU_D}{dI_D} = \frac{KT}{q_0 I_D} \simeq \frac{1}{40 I_D}$$ $I_D(mA) \Rightarrow R_D$

1) P.S.F $U_0(V_{cc}, 0)$



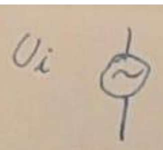
P.p. că D conducție

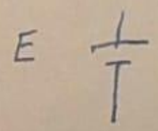
$U_D = 0{,}6 = U_D$

$V_{cc} = U_D + R_1 I_D \Rightarrow$

$$I_D = \frac{V_{cc} - U_D}{R_1} \simeq 1mA$$

2) liniarizare $\Rightarrow$ $R_D = \frac{1}{40 I_D} = 25\,\Omega$





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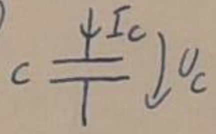


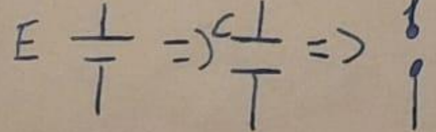
$I_c = C \frac{dU_c}{dt} \Rightarrow$

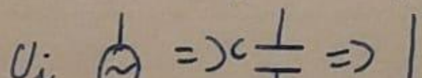
$I_c > 0$, $U_c > 0$, $U_c \nearrow$

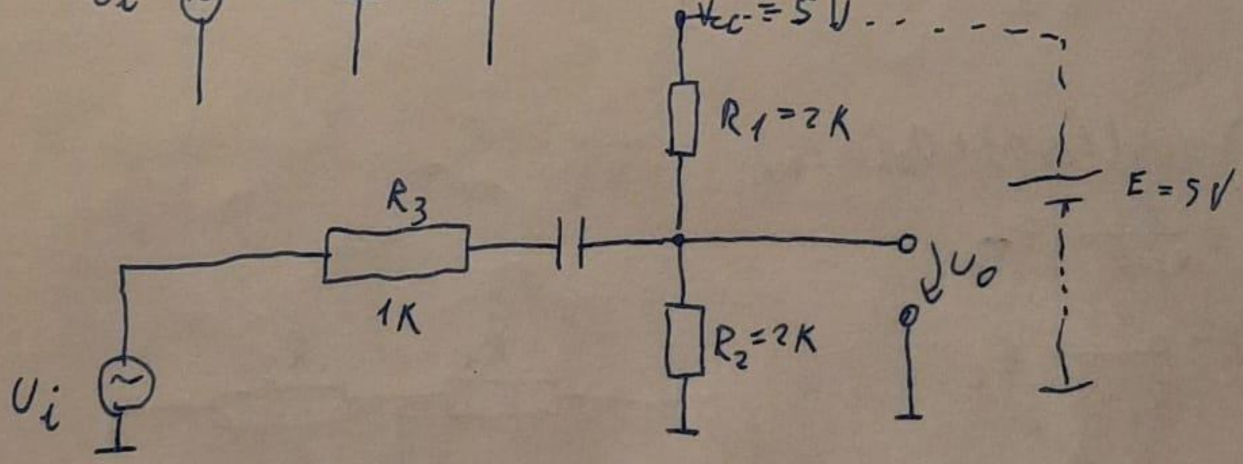
$I_c = 0$, tensiune const., U_c const.

$I_c < 0$, $U_c < 0$; $U_c \searrow$

$$\frac{I_c}{C} = \frac{dU_c}{dt}$$

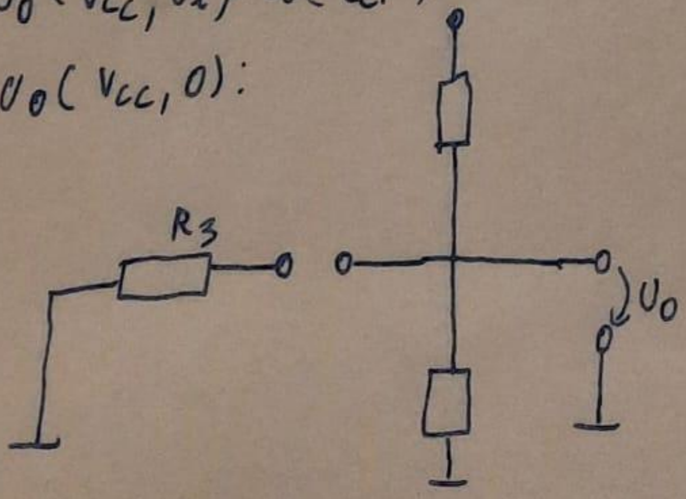




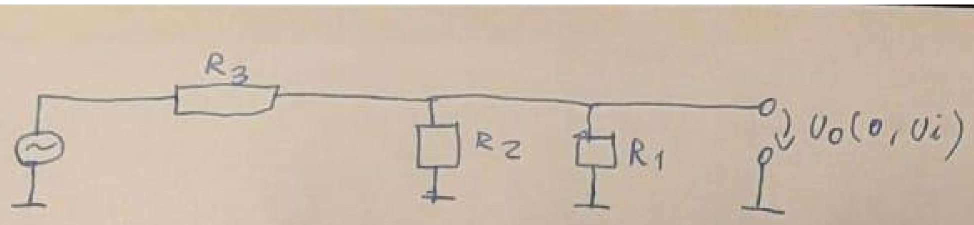

Vcc = 5 V
R1 = 2K
E = 5V
R3
1K
R2 = 2K
U0
Ui

$U_0(V_{cc}, V_i) = U_0(V_{cc}, 0) + U_0(0, V_i)$

$U_0(V_{cc}, 0)$:

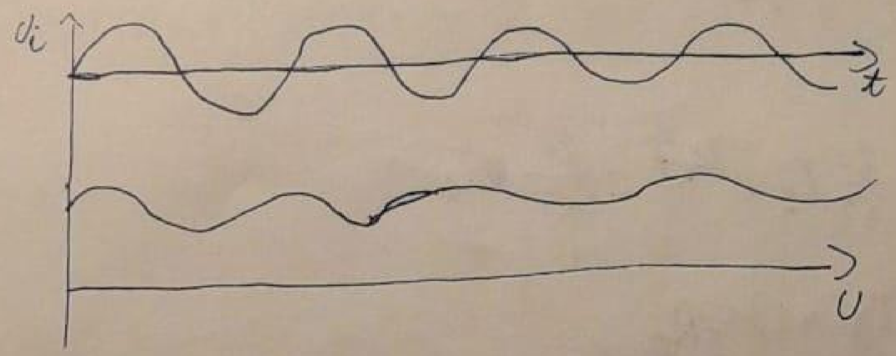


$U_0(V_{cc}, 0) = V_{CC} \frac{R_1}{R_1 + R_2} = 2{,}5\ V$


R3
R2
R1
$U_0(0, U_i)$

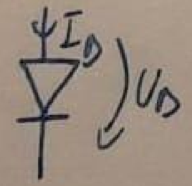
$$U_0(0,U_i) = \frac{R_1 || R_2}{R_3 + R_1 || R_2} U_i = \frac{1K}{1K+1K} U_i = \frac{U_i}{2}$$

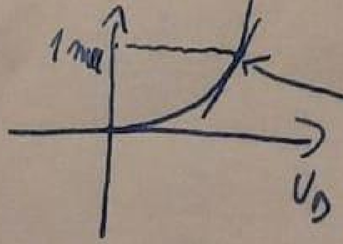
$$U_0(V_{CC}, U_i) = 2{,}5\,V + \frac{U_i}{2}$$



Reactanța condensatorului: $\frac{1}{j\omega C}$

2.





P.S.F. (punct static de funcționare)

Curs 10
DEAEA

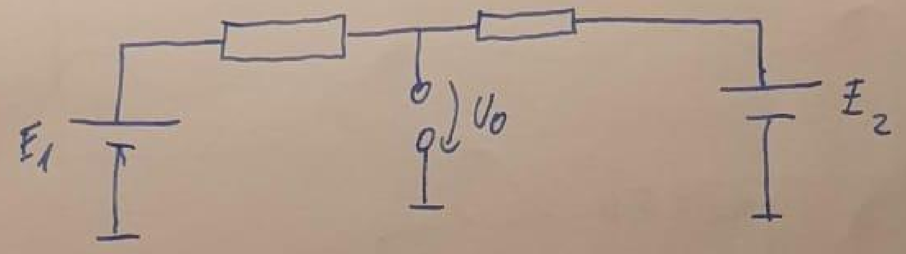
REGIM BINAR

1) - SUPER POZITIE

dacă cir. liniar

funcții liniară $f(\lambda a + \beta b) = \lambda f(a) + \beta f(b)$

$f(u_1, u_2) = f(u_1, 0) + f(0, u_2$

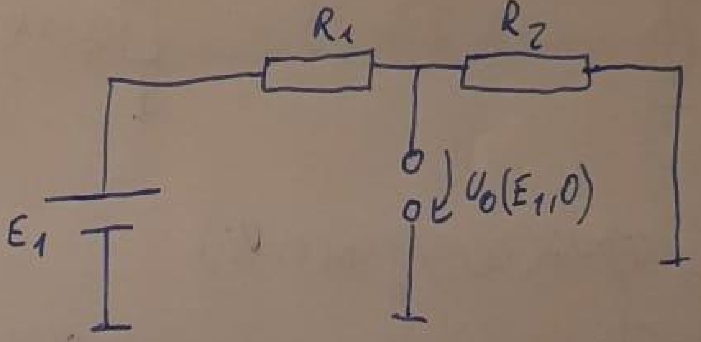


$$U_0(E_1, E_2) = U_0(E_0, 0) + U_0(0, E_2)$$

$$U_0(E_1, 0) = \frac{R_2}{R_1+R_2} \cdot E_{\cancel{2}1}$$

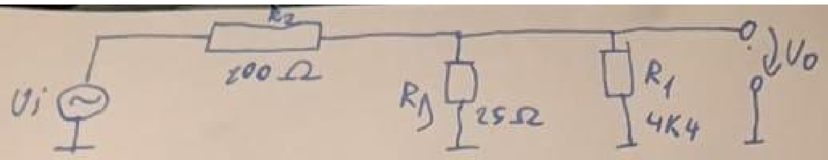
$$U_0(0, E_2) = \frac{R_1}{R_1+R_2} E_2$$

$$\cancel{U_0(E_1, E_2)} = \frac{R_2 E_1}{R_1+R_2} + \cancel{\frac{R_1 E_2}{}}$$



$$U_0(E_1, E_2) = \frac{R_2 E_1 + R_1 E_2}{R_1 + R_2} =$$

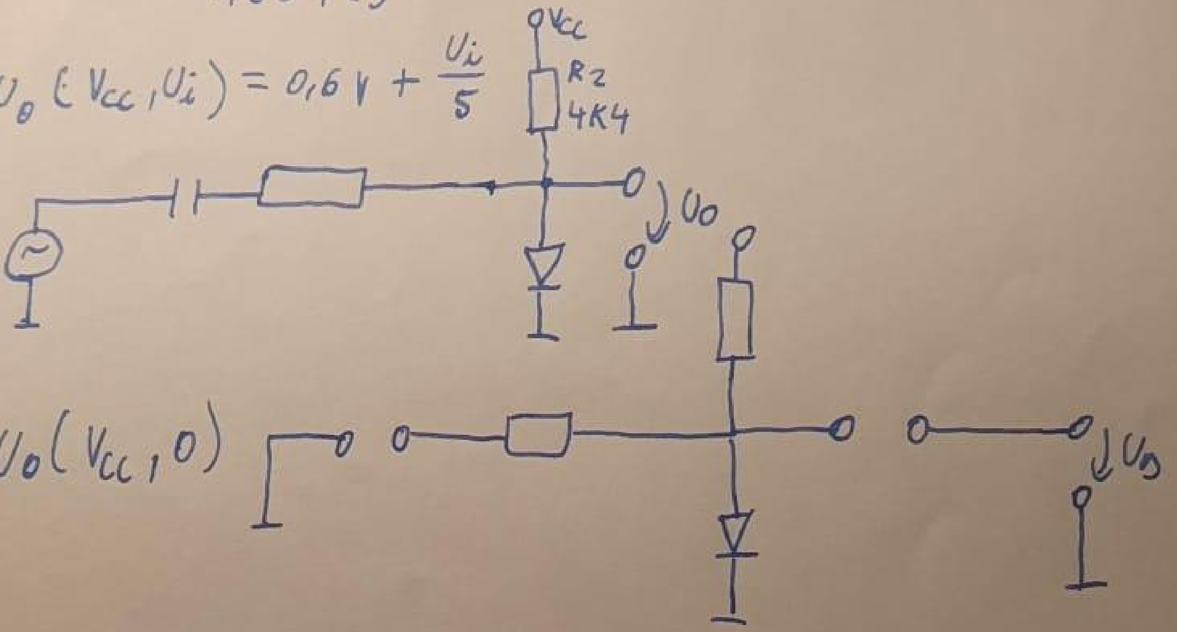
$$= \frac{\frac{E_1}{R_2} + \frac{E_2}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}}$$


R2
100 Ω
Ui
RD
25Ω
R1
4K4
Uo

$$U_o(0,U_i) = \frac{R_1 \parallel R_D}{R_1 \parallel R_D + R_2} U_i \simeq \frac{25\,\Omega}{25+100\,\Omega} U_i = \frac{U_i}{5}$$

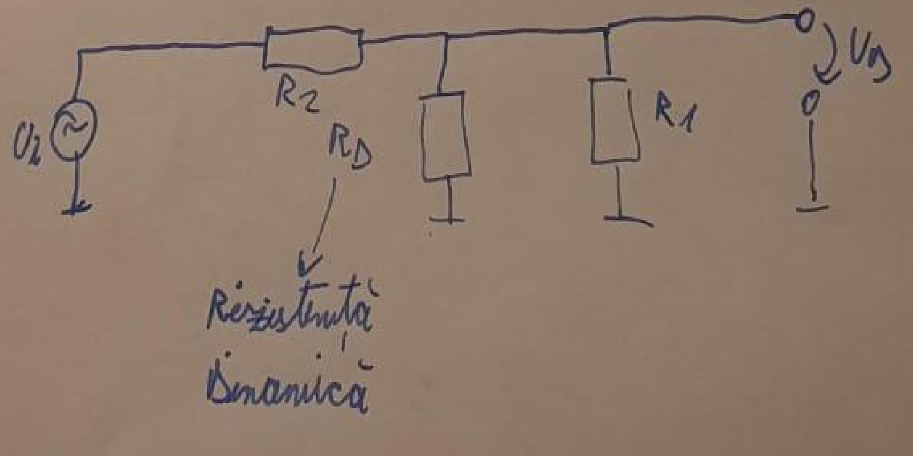
$$R_1 \parallel R_D = \frac{25 \cdot 4400}{4400+25} \simeq 25$$

$$U_o(V_{cc},U_i) = 0{,}6\,V + \frac{U_i}{5}$$


Vcc
R2
4K4
Uo
Uo

$U_o(V_{cc},0)$

$U_o(0,U_i)$ nu se schimbă nimic


Ui
R2
RD
R1
Uo
Rezistență dinamică

$$U_o(V_{cc},U_i) = \frac{U_i}{5}$$